

Urvashi

PhD Researcher · Chemical Engineering & Computational Modelling

📍 Swansea University, United Kingdom

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About

PhD researcher combining **density functional theory**, **quantum simulation** and **experimental materials characterisation** to investigate energy-storage materials, with a focus on sodium-ion conducting solid-state electrolytes. Experienced with Quantum ESPRESSO, Qiskit and scientific Python, with a strong record of interdisciplinary collaboration and peer-reviewed research.

Education

PhD in Chemical Engineering (Energy Innovation)
Oct 2025 – Present

Swansea University, United Kingdom

Advanced materials for energy storage and conversion, integrating atomistic modelling with experimental characterisation.

MSc Physics
🏅 Gold Medalist

2022 – 2024

University of Allahabad, India · CGPA 9.96/10

Specialisation: Laser and Spectroscopy.

BSc in Mathematics, Physics and Chemistry
2019 – 2022

University of Allahabad, India · 90.59%

Research Experience

PhD Researcher
Oct 2025 – Present

Swansea University, United Kingdom

- Investigating materials for energy storage and conversion through atomistic modelling and experimental characterisation.
- Using density functional theory (DFT) to study electronic structure and materials properties.
- Applying scientific programming and numerical methods to support computational materials research.

Junior Research Fellow
Jul 2024 – Aug 2025

Department of Physics, Indian Institute of Technology Roorkee · Supervisor: Prof. P. C. Srivastava

- Investigated ground and excited states of ^{18}O and ^{58}Ni using variational quantum eigensolver methods for nuclear many-body systems.
- Developed and tested workflows in Qiskit, Python and Fortran; mapped shell-model Hamiltonians to qubits using the Jordan–Wigner transformation.
- Applied unitary coupled-cluster ansätze and benchmarked quantum results against conventional shell-model predictions for convergence and accuracy.
- Completed advanced coursework in nuclear physics, numerical analysis, computational nuclear physics and quantum computing (CGPA 9.444/10).

Publications

1 **Melt Quenching Process for the Synthesis of $Na_5MSi_4O_{12}$-Type Compounds, Sodium-Ion Conducting Solid Electrolytes**

A. Michalak, **Urvashi**, M. A. Reddy

ACS Omega (2026) · doi:10.1021/acsomega.6c08181

2 **Synthesis, structure, stability, and ionic conductivity of $Na_5MSi_4O_{12}$ type solid electrolytes for solid-state sodium batteries**

A. Michalak, **Urvashi**, T. O. Dunlop, O. Clemens, A. R. Munnangi

EES Batteries (2026) · doi:10.1039/D6EB00059B

3 **Emerging trends and structure–activity insights of Schiff base–lanthanide complexes as antibacterial agents**
S. Paswan, N. Jaiswal, **Urvashi**, A. Nath, M. Kumar, D. Mishra, N. Yadav
RSC Advances 16 (2026), 3478–3508 · doi:10.1039/D5RA08181E

Technical Expertise

COMPUTATIONAL METHODS
Density functional theory · Electronic structure · Phonons · CI-NEB · Quantum computing · Machine learning

PROGRAMMING
Python · Qiskit · Fortran · C++ · Bash / shell · Java

SCIENTIFIC SOFTWARE
Quantum ESPRESSO · ORCA · NuShellX · Gnuplot · Origin · GaussView · Avogadro · SRIM · LaTeX

EXPERIMENTAL METHODS
XRD · XPS · Raman · SEM · Impedance spectroscopy · FTIR

PRODUCTIVITY
Git & GitHub · Linux / HPC workflows · Excel · Word · PowerPoint

Awards & Achievements

2024 **Ward Vidyant Memorial Gold Medal**, University of Allahabad, for outstanding academic performance
2024 **Ranjana and Hari Prakash Gold Medal**, for the highest marks in MSc Physics (2022–2024)
2024 **First in order of merit**, MSc Physics, University of Allahabad
2024 **MHRD fellowship** for PhD study as a GATE-qualified candidate
2022 **All India Rank 1**, MSc Physics entrance examination, University of Allahabad
2022 **IIT JAM** qualified

Conferences & Training

2025 DTP-TALENT Summer School: Quantum Computing for Nuclear Physics, ECT*, Trento, Italy
2025 Workshop on Quantum Technology, IIT Roorkee
2024 68th DAE Symposium on Nuclear Physics, IIT Roorkee
2023 Meghnad Saha Memorial International Conference on Frontiers of Physics, University of Allahabad

Languages

🇮🇳 **Hindi**: native
🇬🇧 **English**: professional working proficiency (IELTS Academic, CEFR B2)
🏴 **Welsh**: beginner

Projects & Outreach

🧪 Lab Tools
Streamlit app with calculators and notes for materials synthesis, XRD, spectroscopy, impedance and battery work.

🎓 Attack on Scholar
Co-founder. Tutorials and beginner-friendly lectures (like HPC) for early-career researchers, on the web and YouTube.